

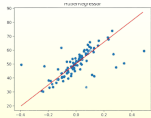
Data Collection



$$\mathcal{D} = \{I_1, I_2, \dots, I_N\}$$

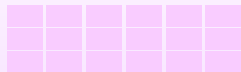


Data Preprocessing



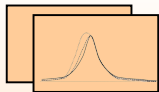
$$I_t = \{I_{t-1}, \text{ if valid}\}$$

Resample Data


 T_s
 f_s


$$I_r = I_{T_s r}$$

EWMA filtering



$$X'_t = \alpha L_t + (1 - \alpha) L'_{t-1}$$

Temporal Encoding

$$\{I'_1, I'_2, I'_3, \dots, I'_r\}$$

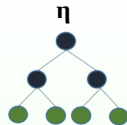
$$x_k = [I'_{k-w}, I'_{k-w+1}, \dots, I'_{k-1}]$$



w-sliding buffer

 X_k

LGBM Regression



$LightGBM_1$

$LightGBM_n$

$LightGBM_N$

$$y_k = \sum_{m=1}^M \eta f_m(x_k)$$